# Huawei Access Node HEB-ACC-75 Minor Failure Resolution Procedure for Hardware/Environmental Faults (ASIC Memory Parity Errors, Line Card Crashes, Power Supply Redundancy Loss)

**Node:** HEB-ACC-75 | **Vendor:** Huawei | **Severity:** Minor | **Alarm Code:** ALARM_4558

## Description

This Standard Operating Procedure (SOP) outlines the diagnostic and resolution steps for minor failures on Huawei Access Node HEB-ACC-75 at Hebbal Edge site, primarily attributed to Hardware/Environmental Faults (ASIC memory parity errors, line card crashes, and power supply redundancy loss). The Huawei Access Node plays a pivotal role in the access network domain, providing services such as broadband access, IP transmission, and Ethernet services. The device's architecture comprises multiple line cards, each equipped with an Application-Specific Integrated Circuit (ASIC) that manages data processing and forwarding. The ASIC is responsible for enforcing Quality of Service (QoS) policies, implementing traffic shaping and policing, and performing other critical functions. If an ASIC experiences memory parity errors, it can lead to line card crashes and power supply redundancy loss, resulting in service outages. This SOP aims to assist in identifying the root cause of the issue and performing the necessary steps to restore services.

## Pre-requisite Checks

- Verify that all necessary permissions and approvals have been obtained before proceeding.

- Ensure that the Hebbal Edge site is physically accessible, and all required personnel have been dispatched.

- Check the status of the power supply redundancy and ensure that the backup system is functioning correctly.

- Verify that the redundant link status is normal and can be used for failover purposes.

## Potential Root Causes

| Cause | Probability |
|---|---|
| ASIC Memory Parity Errors | High |
| Line Card Crashes | High |
| Power Supply Redundancy Loss | Medium |

## Diagnostic Commands & Expected Outputs

```
Command: display device, Expected Output: Device information,
including ASIC status and line card details., Description: Displays
```

```
the current device information, including ASIC status and line card details. This command helps isolate the issue to a specific ASIC or line card.

Command: display power, Expected Output: Power supply status, including redundancy and backup information., Description: Displays the current power supply status, including redundancy and backup information. This command helps verify the power supply redundancy and ensure that the backup system is functioning correctly.

Command: display redundancy, Expected Output: Redundancy status, including link and power supply information., Description: Displays the current redundancy status, including link and power supply information. This command helps verify the redundant link status and ensure that it can be used for failover purposes.
```

## Resolution Steps

Step 1: Step: Replace the faulty ASIC or line card., Command: replace , Description: Replaces the faulty ASIC or line card with a new one. This step is critical in restoring services and preventing further outages.

Step 2: Step: Verify power supply redundancy., Command: verify power redundancy, Description: Verifies that the power supply redundancy is functioning correctly and that the backup system is available for failover purposes.

Step 3: Step: Verify redundant link status., Command: verify redundant link, Description: Verifies that the redundant link status is normal and can be used for failover purposes.

## Post-Check Verification

```
Command: display device, Expected Output: Device information, including ASIC status and line card details., Description: Verifies that the ASIC and line card details are correct and that services are restored.

Command: display power, Expected Output: Power supply status, including redundancy and backup information., Description: Verifies that the power supply redundancy and backup information are correct and that services are restored.

Command: display redundancy, Expected Output: Redundancy status, including link and power supply information., Description: Verifies that the redundant link status and power supply redundancy are correct and that services are restored.
```

## Roll-back Procedures

- Step: Restore the original configuration., Command: undo replace , Description: Restores the original configuration and replaces the new ASIC or line card with the original faulty one.

- Step: Verify the original configuration., Command: display device, Description: Verifies that the original configuration is correct and that services are restored.